

ITAI-1371 - Intro to Machine Learning
Reflective Journal

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This lab helped me understand how the stages of a machine learning workflow connect. Loading data, selecting features, splitting the dataset, training models, and evaluating predictions each serve a purpose. Following the Wine classification example made these connections more concrete and showed me why a working program still requires careful interpretation.

The distinction between learning types became clearer when I considered the feedback available during training. Supervised learning uses labeled examples, such as wine measurements paired with their correct classes. Unsupervised learning searches for patterns without known answers. Reinforcement learning involves an agent interacting with an environment and learning through rewards or penalties. Choosing an approach depends on the problem and the available data.

The model comparison showed me that results must be interpreted within the experiment's limits. Logistic Regression achieved 88.9% accuracy, while the Decision Tree achieved 83.3%. This difference represented only two predictions among 36 test samples. Logistic Regression performed better in this test, but that does not establish that it will always be the better model. The provided three-feature experiment also achieved 83.3%, demonstrating that changing the feature combination does not automatically improve performance.

My greatest challenge was understanding precision and recall beyond overall accuracy. Reviewing the classification report alongside the confusion matrix helped me distinguish them. For class_2, Logistic Regression identified seven of ten actual samples, giving a recall of 0.70. Seven of its eight predictions of class_2 were correct, giving a precision of approximately 0.88. These results showed me that a model can have good overall accuracy while still missing several examples of a particular class.

I connected this workflow to English language education. A supervised model could help identify students who may need additional practice using assessment results and learning records paired with teacher reviewed labels. Teachers would still need to review its recommendations, protect student privacy, and consider fairness.

In future experiments, I would explore feature scaling and compare feature combinations through cross-validation on the training data. I would reserve the test set for final evaluation. My main takeaway is that reliable machine learning requires understanding both the workflow and the mistakes behind the final score.

Danh Nguyen

My Understanding of Machine Learning Types

Supervised Learning: I understand supervised learning as learning from examples that already have the correct answers. The model uses those examples to make predictions on new data.

Unsupervised Learning: This is when a model looks for patterns or groups in data without being told what the answers should be.

Reinforcement Learning: This is learning by trying different actions and getting feedback. The agent gradually learns which actions lead to better rewards.

My Analysis of the Wine Classification Project

Best performing model: Logistic Regression performed best, with 88.9% accuracy. The Decision Tree reached 83.3%.

Why do you think this model performed better?

In my opinion Logistic Regression worked better with the features we selected. The Decision Tree was limited to a depth of three, so it might not have picked up enough detail.

What would you try next to improve performance? I would try scaling the data and adding more features. My three-feature model scored 83.3%, so changing the features did not improve the result in this experiment.

Real-World Application Ideas

Industry of Interest: Cybersecurity.

ML Problem: Identifying phishing emails.

Type of ML: Supervised learning.

Data Needed: Examples of phishing and legitimate emails, including their messages, sender details, and links.

Key Learnings

Most important concept learned: I learned why we separate training and testing data. A model needs to make good predictions on new examples, not just the ones it learned from.

Most challenging part: The hardest part for me was understanding what precision and recall tell us beyond accuracy.

Questions for further exploration: I want to know how to choose better features and tell when a model is overfitting.

Great Inioluwa Olayiwola

My Understanding of Machine Learning Types

Supervised Learning: You train a model with labeled examples. For each example, you give the input features (x) and the correct output (y). The model then studies the link between x and y. After that, it uses what it learned to produce predictions for new, unseen inputs.

Unsupervised Learning: Looking for patterns, structures, or clusters in the input data (x) by itself. This does it without any fixed labels or target outputs (y).

Reinforcement Learning: You can train an agent by letting it try things and learn from the results. The agent takes actions in an environment, then it gets feedback. The goal is to end up with higher total rewards over time and fewer penalties along the way.

My Analysis of the Wine Classification Project

Best performing model: The best performing label was Logistic Regression. It's accuracy is 88.9%

Why do you think this model performed better?: Logistic Regression performed better because the relationship between the four selected features (alcohol, malic_acid, ash, and alcalinity_of_ash) and the wine classes is relatively linear. Decision Trees with a strict depth constraint (max_depth=3) can create coarse, axis-aligned decision boundaries that struggle with fine numerical boundaries. Furthermore, Logistic Regression achieved perfect precision and recall (1.00) for class_0 and only misclassified 4 samples in total (1 in class_1 and 3 in class_2).

What would you try next to improve performance?

Feature Scaling: Apply StandardScaler to normalize features, especially when including high-magnitude variables like proline.

Feature Selection & Expansion: Use all 13 features instead of limiting the inputs to 3 or 4, or apply feature selection techniques like SelectKBest.

Hyperparameter Tuning: Experiment with different max_depth settings or criterion types for the Decision Tree, or tune the regularization strength parameter (C) in Logistic Regression.

Real-World Application Ideas

Industry of Interest: Healthcare / Medical Diagnostics

ML Problem: Automated detection and classification of cardiac arrhythmias from patient ECG monitoring.

Type of ML: Supervised Learning

Data Needed: Time-series electrocardiogram (ECG) heart rate signal data paired with verified diagnostic annotations provided by cardiologists.

Key Learnings

Most important concept learned: I learned to split the data into a training part (80%) and a test part (20%). I used stratification in `train_test_split` so the test set stays representative. This helps me judge how the model will do on new, unseen data and it reduces overfitting.

Most challenging part: Picking the right features was challenging for me. For instance, I moved from the original 4 features, which gave 88.9 percent accuracy, to a new set of three: ['alcohol', 'color_intensity', 'proline']. After that change, Logistic Regression dropped to 83.3 percent accuracy.

Questions for further exploration: What is the best way to deal with class imbalance when the classes are not evenly spread? Also, what steps can stop model quality from falling off once the model is used on real data that comes in as a steady stream?

Karima Sadruddin

Understanding the learning approaches

This lab helped me separate the three main machine learning approaches by asking what feedback the model receives. In supervised learning, the training examples include labels or correct answers. In the Wine dataset, chemical measurements were the input features and the wine cultivar was the target. Unsupervised learning instead searches for structure in data without known labels, such as grouping customers by similar behavior. Reinforcement learning is different because an agent interacts with an environment and learns from rewards or penalties. These approaches are not interchangeable; the problem and available feedback determine which one makes sense.

How the workflow fits together

The workflow made more sense when I viewed its stages as connected decisions rather than isolated tasks. First, the problem and data must be understood. Exploratory data analysis can reveal class imbalance, feature relationships, and possible quality issues. Then features and a target are selected, and the dataset is split into training and testing portions. Training teaches the algorithm from the training data, while evaluation checks predictions against examples the model did not train on. The confusion matrix adds detail by showing which classes were predicted correctly and where errors occurred. This is important because accuracy alone can hide the types of mistakes a model makes.

Model comparison and limitations

Comparing logistic regression with a decision tree showed that different algorithms can learn different decision patterns from the same data. A model that performs better on one test split is not automatically best for every future dataset. The chosen features, data split, model settings, and evaluation metric all affect the result. To improve confidence, I would use cross-validation to compare models, consider feature scaling for logistic regression, test other feature combinations, and examine precision and recall alongside accuracy. I would also keep the test set separate until the final evaluation to reduce the chance of tuning decisions to the test results.

Reflection and next questions

One broader lesson is that machine learning quality depends on the whole process, not just the algorithm. Data quality, appropriate validation, and careful interpretation are necessary before a model could be used responsibly. In an education example, a supervised model might identify students who could benefit from extra support using attendance and assignment patterns, but privacy, fairness, and human review would need to be considered. Questions I would like to explore further include how cross-validation changes confidence in a model comparison, how to identify the features most responsible for predictions, and how to monitor a deployed model when real-world data changes.

Quan Thach

When I first opened the Module 3 lab, I expected it to be mostly code. It turned out to be a nice mix of ideas and practice, and by the end I understood the bigger picture of machine learning much better. The first thing I learned was the three types of learning. Supervised learning uses labeled examples, so the model has both the inputs and the correct answers while it trains. Unsupervised learning has no answers, so the model looks for hidden patterns on its own, like grouping customers by behavior. Reinforcement learning is learning by trial and error, where an agent gets rewards or penalties. The lab said we would focus mostly on supervised learning, which made the course feel more manageable.

The part I enjoyed most was walking through the workflow with the Wine dataset. Before this, building a model sounded complicated to me, but the lab broke it into small steps. I explored the data first, checking class balance, missing values, and correlations.

Then I picked features, split the data into training and testing sets, trained a Logistic Regression model and a Decision Tree, and compared them. The idea of splitting the data finally made sense to me: if you only test on what the model has already seen, you can't tell whether it is truly learned or just memorized. I also liked that the lab didn't stop at accuracy. The classification report and confusion matrix showed me exactly where a model gets things right and where it makes mistakes.

Another useful section was about data types. I had never really thought about the difference between continuous numbers, categories, ordinal values, text, and true or false data. The lab made it clear that each type needs different preprocessing and different algorithms. That helped me see why real projects need so much care before any model is trained. The hands-on task pushed me to experiment myself. I chose flavonoids, color intensity, and proline as my features. I noticed that proline has values in the hundreds while the others are small, so I added scaling. I also used cross-validation because a small test set can make a single accuracy score misleading. It was a small change, but it taught me that good results depend on thoughtful choices, not only on running the code.

The assessment questions were a quick way to check myself. Matching each scenario to its learning type showed me that keywords like "labeled," "discover," and "rewards" really do point to the right answer. The real-world case studies on recommendation systems, fraud detection, and medical diagnosis also helped. They showed that the same workflow behind a simple wine model is used in serious, everyday systems. When I thought about applications for transportation, I could see how self-driving cars and route planning fit into what I had just learned.

The most challenging part for me was understanding how many small decisions go into one model, such as which features to use, how to evaluate it, and how to know if it can be trusted. The final checklist and lifecycle helped because they gave me a clear order to follow. My biggest takeaway is to start simple, understand the data first, always test on unseen data, and keep improving step by step. Next, I want to try the practice ideas from the lab, like using

other datasets and applying the same workflow to a new problem. This lab made machine learning feel less intimidating and a lot more like something I can actually learn.